

**1. Sequence retrieval and dataset assembly**  
PANTHER · OrthoDB · UniProt · NCBI BLAST (Metazoa) + EukProt (Amoebozoa)  
domain-architecture check (UniProt/InterPro) · balancing tree (MEGA12)  
scope check: EukProt-wide + PhylomeDB (choanoflagellates, other opisthokonts)

**2. Similarity searches**  
PSI-BLAST (NCBI nr + EBI UniProtKB), seeded with H. sapiens Q96JY0  
filtered by the same domain-architecture check as above

81-taxon dataset  
(72 Metazoa + 9 Amoebozoa)

**3. Alignment and motif ID**  
MAFFT L-INS-i  
↓  
full-length alignment (1288 col)  
domain-only alignment (336 col)  
ECHC/DEDD verified vs H. sapiens

**4. Structural comparison**  
AlphaFold DB structures  
(H. sapiens, D. melanogaster,  
A. castellanii)  
PyMOL superposition · Foldseek search

**5. Phylogenetic analysis**  
IQ-TREE3 (ModelFinder, UFBoot2,  
SH-aLRT) on both alignments,  
rerooted on Amoebozoa

RMSD comparisons  
(structural homology)

full-length tree + domain-only tree

**Results**  
which tree is primary; motif conservation  
across taxa; structural homology; HMG-box